

DSC 206: Homework 3

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Assignment 1 Consider the two points $\mathbf{x} = (1, 0, 1, 0, 1, 0)$ and $\mathbf{y} = (0, 1, 1, 0, 1, 0)$. Compute the distance between them under the L_0 (Hamming), L_1 , L_2 (Euclidean), and L_∞ distances.

Solution. The coordinate-wise differences are $\mathbf{x} - \mathbf{y} = (1, -1, 0, 0, 0, 0)$, so $\mathbf{x}$ and $\mathbf{y}$ disagree in exactly the first two coordinates.

$$L_0(\mathbf{x}, \mathbf{y}) = \#\{i : x_i \neq y_i\} = 2,$$

$$L_1(\mathbf{x}, \mathbf{y}) = \sum_i |x_i - y_i| = 1 + 1 = 2,$$

$$L_2(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_i (x_i - y_i)^2} = \sqrt{2},$$

$$L_\infty(\mathbf{x}, \mathbf{y}) = \max_i |x_i - y_i| = 1.$$

Assignment 2 On the space of nonnegative integers, decide whether each of the following is a distance measure. Recall the axioms: (i) $d(x, y) \geq 0$, (ii) $d(x, y) = 0 \iff x = y$, (iii) $d(x, y) = d(y, x)$, (iv) triangle inequality $d(x, z) \leq d(x, y) + d(y, z)$.

(a) $\max(x, y)$ — the larger of x and y ;

(b) $\text{diff}(x, y) = |x - y|$;

(c) $\text{sum}(x, y) = x + y$.

Solution. (a) $\max(x, y)$ is **not** a distance. It violates axiom (ii): for any $x > 0$, $\max(x, x) = x \neq 0$, so $d(x, x) \neq 0$. Concretely, $d(3, 3) = 3$.

(b) $\text{diff}(x, y) = |x - y|$ is a distance. This is the restriction of the standard Euclidean metric on $\mathbb{R}$ to $\mathbb{N}$.

(i) $|x - y| \geq 0$ by definition of absolute value.

- (ii) $|x - y| = 0 \iff x - y = 0 \iff x = y$.
- (iii) $|x - y| = |-(y - x)| = |y - x|$.
- (iv) $|x - z| = |(x - y) + (y - z)| \leq |x - y| + |y - z|$ by the triangle inequality for $|\cdot|$ on $\mathbb{R}$.

(c) $\text{sum}(x, y) = x + y$ **is not a distance**. It violates axiom (ii): $d(2, 2) = 4 \neq 0$. (It also fails the triangle inequality, e.g. $d(1, 3) = 4 > d(1, 2) + d(2, 3) = 3 + 5 = 8$ does hold here, but the failure of (ii) already rules it out.)

Assignment 3 Starting from the family of minHash functions (so a single hash matches with probability equal to the Jaccard similarity s), describe the effect on the matching probability of the following constructions. Write the probability as a function of s .

- (a) 2-way AND followed by 3-way OR.
- (b) 3-way OR followed by 2-way AND.
- (c) 2-way AND, then 2-way OR, then 2-way AND.
- (d) 2-way OR, then 2-way AND, then 2-way OR, then 2-way AND.

Solution. Recall that an r -way AND on a family with collision probability p produces collision probability p^r , and an r -way OR produces $1 - (1 - p)^r$. We compose stage by stage starting from $p_0 = s$.

- (a) AND-2 gives s^2 ; OR-3 gives

$$p(s) = 1 - (1 - s^2)^3.$$

- (b) OR-3 gives $1 - (1 - s)^3$; AND-2 gives

$$p(s) = (1 - (1 - s)^3)^2.$$

- (c) AND-2 gives s^2 ; OR-2 gives $1 - (1 - s^2)^2$; AND-2 gives

$$p(s) = (1 - (1 - s^2)^2)^2.$$

- (d) OR-2 gives $1 - (1 - s)^2 = 2s - s^2$; AND-2 gives $(2s - s^2)^2 = s^2(2 - s)^2$; OR-2 gives $1 - (1 - s^2(2 - s)^2)^2$; AND-2 gives

$$p(s) = \left(1 - (1 - s^2(2 - s)^2)^2\right)^2.$$

Assignment 4 Consider a Bloom filter with $n = 8$ bits of memory, a set S of $m = 100$ allowed keys, and $k = 3$ independent random hash functions. Compute (i) the probability that an element of S is blocked, and (ii) the probability that an element not in S is blocked. Do not use the simplified large- n approximation.

Solution. (i) Element in S . A Bloom filter has *no false negatives*: every bit set during the insertion of $x \in S$ remains set, so on a query for x all k probed bits are 1 and the filter accepts. Hence

$$\Pr[\text{block} \mid x \in S] = 0.$$

(ii) Element not in S . After inserting the m keys, $km = 300$ independent uniform throws have been made into the $n = 8$ bins. Let B denote the (random) set of distinct bins hit; bit i equals 1 iff $i \in B$.

For a query $x \notin S$, let $H = (h_1(x), \dots, h_k(x))$ be its k hash images and let $T(H) = \{h_1(x), \dots, h_k(x)\}$ be the set of distinct probed positions, with $t = |T(H)|$. The filter blocks x iff every bit in $T(H)$ is 1, i.e. $T(H) \subseteq B$.

Conditional on $T(H) = T$ with $|T| = t$, the event $\{T \subseteq B\}$ says that none of the t specified bins is missed by all km insertions. By inclusion–exclusion,

$$\Pr[T \subseteq B] = \sum_{j=0}^t (-1)^j \binom{t}{j} \left(\frac{n-j}{n} \right)^{km} =: f(t).$$

The number of ordered k -tuples from $\{1, \dots, n\}$ that have exactly t distinct values is $N(k, t) = \binom{n}{t} t! S(k, t)$, where $S(k, t)$ is the Stirling number of the second kind. Since each of the k query hashes is uniform on $\{1, \dots, n\}$,

$$\Pr[\text{block} \mid x \notin S] = \frac{1}{n^k} \sum_{t=1}^{\min(k, n)} N(k, t) f(t).$$

For $k = 3$ we have $N(3, 1) = n$, $N(3, 2) = 3n(n-1)$, and $N(3, 3) = n(n-1)(n-2)$ (which sum to n^3). With $n = 8$, $km = 300$ this becomes

$$\Pr[\text{FP}] = \frac{1}{512} [8f(1) + 168f(2) + 336f(3)],$$

where

$$\begin{aligned} f(1) &= 1 - \left(\frac{7}{8}\right)^{300}, \\ f(2) &= 1 - 2\left(\frac{7}{8}\right)^{300} + \left(\frac{6}{8}\right)^{300}, \\ f(3) &= 1 - 3\left(\frac{7}{8}\right)^{300} + 3\left(\frac{6}{8}\right)^{300} - \left(\frac{5}{8}\right)^{300}. \end{aligned}$$

Numerically, $(7/8)^{300} = e^{300 \ln(7/8)} \approx e^{-40.06} \approx 4.4 \times 10^{-18}$, and the other two are vastly smaller. Thus $f(1), f(2), f(3)$ all equal 1 to extraordinary precision, giving

$$\Pr[\text{block} \mid x \notin S] \approx \frac{8 + 168 + 336}{512} = \frac{512}{512} = 1.$$

With only 8 bits storing 100 keys via 3 hashes, all bits are essentially certain to be set and the filter rejects almost nothing.

Assignment 5 Consider the data stream

1, 2, 6, 3, 2, 2, 3, 1, 6, 6.

- (a) What is the number of distinct items?
- (b) Run the Flajolet–Martin algorithm with alphabet size $m = 6$ and hash $h(x) = 3x + 1 \bmod 7$. Show your work.
- (c) Find the frequency of each element.

Solution. (a) **Distinct count.** The distinct values appearing in the stream are $\{1, 2, 3, 6\}$, so the true number of distinct items is 4.

(b) **Flajolet–Martin.** Apply the hash to each distinct stream value (repeats produce no new information). Since h takes values in $\{0, 1, \dots, 6\}$, we represent each value with $\lceil \log_2 7 \rceil = 3$ bits and let $r(x)$ denote the number of trailing zeros in the binary expansion of $h(x)$.

x	$h(x) = 3x + 1 \bmod 7$	binary (3 bits)	$r(h(x))$
1	4	100	2
2	0	000	3
3	3	011	0
6	5	101	0

We use the convention that $h(x) = 0$ contributes the maximum tail-length $\lceil \log_2 7 \rceil = 3$. The algorithm tracks $R = \max_x r(h(x))$ as the stream is processed; tracing through 1, 2, 6, 3, 2, 2, 3, 1, 6, 6 the running maximum becomes 2, 3, 3, 3, 3, 3, 3, 3, 3, 3. Hence $R = 3$, and the FM estimate is

$$\hat{D} = 2^R = 2^3 = 8.$$

This overestimates the true value 4, which is typical for a single FM register on a tiny stream — in practice one averages many independent hash functions to reduce variance. (If one instead drops the $h = 0$ value as ill-defined, then $R = 2$ and $\hat{D} = 4$, matching the truth exactly.)

(c) **Frequencies.** The stream has length 10. Counting occurrences,

$$f(1) = \frac{2}{10} = 0.2, \quad f(2) = \frac{3}{10} = 0.3, \quad f(3) = \frac{2}{10} = 0.2, \quad f(6) = \frac{3}{10} = 0.3.$$